

Discrepancies in Mobile App Permissions: Exploring Transparency and User Awareness in the Android Ecosystem

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Abstract

Transparency in mobile app permissions is essential for maintaining user trust and protecting privacy as individuals increasingly rely on digital solutions. Mobile applications frequently request access to phone resources, including photos, location data, and contacts upon installation. This study investigates discrepancies in Android app permissions by analyzing 200 applications and conducting a survey representing a diverse demographic group. The analysis focuses on identifying the discrepancies between permissions requested by the apps and those disclosed to users in the Play Store description, aiming to understand the implications of these discrepancies on user trust and privacy. To explore user awareness and attitudes toward permission discrepancies, a mixed-methods survey was conducted with 110 technically proficient participants. The survey findings reveal that, despite being the most technologically proficient demographic, their familiarity and attention with app permissions is only moderate.

CCS Concepts

• **Security and privacy** → **Security services; Access control; Privacy-preserving protocols; Software and application security;**

Keywords

App permissions, privacy awareness, transparency, permission discrepancies, reverse engineering

ACM Reference Format:

Bharathi Donku, Shahriar Rahman Khan, Tariqul Islam, and Raiful Hasan. 2025. Discrepancies in Mobile App Permissions: Exploring Transparency and User Awareness in the Android Ecosystem. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3706599.3719902>

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CHI EA '25, Yokohama, Japan

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ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3719902>

April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 8 pages.
<https://doi.org/10.1145/3706599.3719902>

1 Introduction

Mobile applications now play a pivotal role across various domains including communication, entertainment, productivity, and health-care. As of January 2025, data suggests that 70.3% of the global population utilizes mobile phones, and 67.5% have internet access, highlighting an increasing reliance on these technologies [12]. Moreover, recent findings indicate that the average smartphone user engages with approximately 10 apps daily and 30 apps monthly [8], illustrating the deepening integration of these technologies into everyday life. Given this growing dependency, the need for clarity and transparency in app permissions is now more critical than ever. Mobile apps require access to various device features and user data, managed through a permissions-based system outlined in the *AndroidManifest.xml* file [13].

Recent developments in the Android app ecosystem underscore the importance of user consent for managing app permissions. The Google Play Store facilitates informed decision-making by outlining key functionalities and permissions. However, this study finds that there are discrepancies between the permissions detailed in app descriptions and those requested by the apps, thereby creating a transparency gap. This problem is increased in web-based Play Store descriptions, which frequently fail to provide complete and accurate permission details, further complicating users' ability to make informed decisions during app installation. This lack of transparency in permission disclosures undermines user trust and increases privacy risks, potentially leading to data breaches and unauthorized data sharing [32].

This study is driven by a series of research questions that aim to clarify key aspects of user interaction with mobile application permissions within the Android ecosystem. The following questions guide our inquiry: **RQ1:** *To what extent do Android applications inconsistently disclose permissions between the *AndroidManifest.xml* file and app descriptions in the Google Play Store?* **RQ2:** *How do familiarity, attention, and awareness of app permissions as described in the Google Play Store and managed through the app permission system vary across different demographic groups?* **RQ3:** *How does user*

behavior vary when encountering unnecessary or mismatched app permissions? To systematically address the posed research questions, this study is segmented into two primary components. The initial phase focuses on RQ1, involving an analysis of 200 Android applications. This analysis includes extracting permissions from their *AndroidManifest.xml* files using reverse engineering techniques and subsequently comparing these permissions with those declared in the Google Play Store descriptions. The second phase targets RQ2 and RQ3 and consists of an empirical study with 110 participants, who are young, educated, and technologically adept. This segment is designed to evaluate the participants' familiarity with app permissions, their awareness of superfluous permissions, and their responses to discrepancies between the permissions requested and those disclosed. This bifurcated approach serves to clarify the current landscape of app permission transparency and evaluates both user engagement and the influence of permissions on user trust. To summarize, our main contributions are highlighted as follows:

- We conduct a detailed analysis of 200 Android applications from the Google Play Store, systematically identifying and quantifying discrepancies between the permissions declared in app descriptions and those explicitly required by the applications as specified in their *AndroidManifest.xml* files.
- Our study reveals significant discrepancies in permission listings both in app-based and web-based Play Store descriptions compared to actual permissions in the applications. Specifically, we found that 51% (N=102) of the apps analyzed from the Google Play Store displayed minor discrepancies, with 0-15% of their required permissions not listed in their app descriptions. In contrast, a significant portion, 9.5% (N=19), exhibited major inconsistencies, with 46-60% of their necessary permissions missing from their listings. We identified higher inconsistencies in web-based descriptions, with 43.5% (N=87) of apps missing 46-60% of their permissions and 19% (N=38) missing 61-75%.
- In a detailed user study involving 110 participants, our investigation uncovered gaps in user engagement with app permissions. Remarkably, despite involving a cohort characterized by high education and technological proficiency, 48.3% of the participants demonstrated no to minimal familiarity with app permissions or discrepancies, and 59.1% admitted to paying little to no attention to permissions during the app installation process.

2 Background and Related Works

Users often express significant concerns about privacy risks associated with their devices' cameras and microphones being accessed by apps, as these features can capture sensitive information [2, 9, 11, 24, 26, 30]. Mobile devices also store vast amounts of sensitive data, such as personal files, contacts, communication logs, and location history, which raise serious privacy concerns [21]. Additionally, users are sensitive about their photos, messages, emails, and contact lists [16]. Despite these concerns, users do not know about the sensors that are active on their devices [21] or the type of data being collected by the applications they use [4]. Permission descriptions are also important for the users to make informed decisions about an app and if they are vague or even misleading, makes it difficult to understand what is truly required [14]. In many

cases, permissions are requested not for the app's core functionality but to support third-party libraries integrated into the app [15] and there might be a discrepancy or inconsistency between the permissions listed in the manifest file and the app description. This lack of clarity leaves users uncertain about the decision of granting access to their data [7]. As a result, assessing active access to a device's sensors and the related privacy risks is a significant challenge. Prior researches revealed significant gaps in addressing transparency and user privacy in mobile app permissions. For instance, AutoPer+ predicts app permissions based on descriptions and APIs but focuses on predictive accuracy rather than the completeness of permission disclosure [18]. Jeon et al. conducted a study to classify apps into risk categories based on permissions but did not compare permissions in manifest files with store descriptions or assess transparency [23]. Another research done by Prange et al. revealed the importance of user awareness and control of privacy permissions on Android apps [31]. These gaps highlight the need for thorough investigations into permission discrepancies and the current situation of user familiarity and awareness about these discrepancies.

Moreover, users often prioritize convenience over privacy, granting sensor access to enable certain features or functionalities without fully understanding the consequences involved [3, 19, 29]. Hayes et al. investigated how mobile apps collect and share personal data without user consent and proposed mitigation strategies like improved transparency and user education to protect privacy [20]. Building upon this, Lavranou et al. analyzed popular mobile app permissions and developed a tool to categorize them, aiming to improve privacy education and support workshops on app privacy [25]. From a user privacy perspective, Liu et al. explored the principal-agent dynamic between app developers and users, emphasizing the need for transparency and trust in the permission-granting process [27]. Despite these advancements, challenges remain in addressing the discrepancy in permission disclosures and user awareness about app permissions [5]. A particularly vulnerable demographic is older adults (65+), who face greater difficulty managing privacy settings and protecting personal information compared to younger users [10, 28]. Research indicates that older individuals struggle with navigating smartphone security features and require tailored support to handle privacy threats effectively [17, 22]. Therefore, user familiarity with permissions, careful review when granting them, and awareness of data privacy risks are crucial in the digital era.

3 Methodology

3.1 Addressing the Discrepancy in Android App Permission Disclosures

Android Application Selection. To get a comprehensive understanding of how apps use permissions in the Android ecosystem, apps were selected across three download ranges: 100–10,000, 100,000–10 million, and over 10 million. This ensured that both popular apps and niche ones were covered. The applications were selected to represent diverse categories, including lifestyle, health and fitness, entertainment, social media, productivity, photography, education, finance, and tools (see Figure 1(a)).

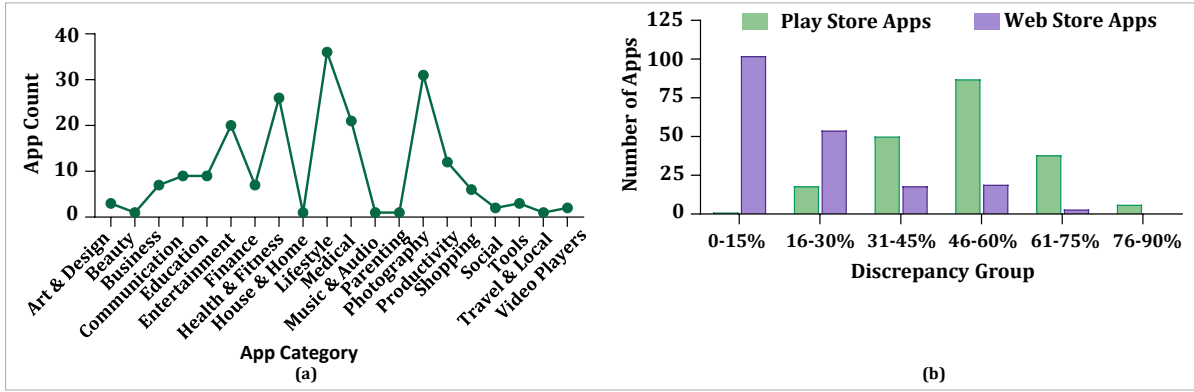


Figure 1: (a) Different categories of 200 selected apps from Google and Web Play Stores (b) Number of apps having permission discrepancy percentage between app description and AndroidManifest.xml file

To keep the data consistent, apps collection was conducted within a specific time-frame from May 11 to May 15, 2024, because developers often update their apps and change permissions. Finally, a comprehensive selection of 200 Android applications was made from the Google Play Store and web store. Lifestyle applications have the highest representation, followed by health and fitness, and entertainment applications, according to the graph. This diverse categorization ensures a broad representation of application types within the Android ecosystem. The systematic selection methodology provided a representative sample of the Android application ecosystem and helped to analyze permission usage patterns across a diverse range of applications.

Data Collection. Data was collected from two distinct sources for each selected application: **Manifest Files:** The permissions were extracted from each application's *AndroidManifest.xml* file using reverse engineering tools like APKTool [33]. To extract permissions, APKTool decodes the APK file, allowing access to the *AndroidManifest.xml* file. This process makes it possible to view and analyze the permissions without needing to install the app. **Google Play Store Descriptions:** Permissions listed in the Google Play Store descriptions were extracted automatically using two methods. For mobile app descriptions, Appium Inspector [1, 6], in combination with Android Studio emulators, was employed to extract permissions from the app's description visible to users before installation. For the web version, the Google Play scraper library was used to extract permissions from the Permissions section of each application's web description, accessible to users on both web and mobile platforms.

Data Comparison and Analysis. Once the app data were collected, the next step involved comparing the permissions listed in the *AndroidManifest.xml* file with those mentioned in the app description on the Google Play Store and Web Permissions. For each app, the total number of permissions requested in the *AndroidManifest.xml* file was compared with the number disclosed in the app description and web permissions. The difference between these two was calculated as a percentage, representing the missing permissions.

3.2 User Study

Design of the Survey Questionnaire. To address the research questions of this study, a comprehensive survey questionnaire (given in Appendix A) was designed and distributed online to a diverse group of participants. The survey consisted of both quantitative and qualitative questions to understand users' familiarity with app permission management, awareness of app permissions, actions taken upon encountering unnecessary or mismatched app permissions, and concerns about user security and privacy through app permissions. The survey included the following sections:

- **Demographics:** These questions (Q1–Q6) given in Appendix A.1 aimed to get information about age, gender, educational background, and app usage frequency of the participants. The survey gathered demographic data to analyze patterns in users' app permission awareness and their behavior when identifying anomalies in app permissions listed in the app description and in app permission manager in their smartphones. It was required for the participants to be smartphone users. The demographic information of the participants are provided in Table 1.

Table 1: Survey Demographics

		N=110	Count	Percentage
Gender	Male		63	57.3%
	Female		47	42.7%
Age	18-24		65	59.1%
	25-34		24	21.8%
	35-44		6	5.5%
	Others		15	13.6%
Education	Bachelor's Degree		61	55.5%
	Master's Degree or Diploma		36	32.7%
	Doctorate Degree		5	4.5%
	Others		8	7.3%
Field of Education	Technology		91	82.7%
	Non-Technology		19	17.3%

- **Smartphone Usage:** These questions (Q7–Q10) given in Appendix A.2 aimed to understand participants' smartphone usage behavior.
- **Familiarity, Awareness, and User Behavior on Permissions:** Questions (Q11–Q30) given in Appendix A.3 aimed to assess users' familiarity with the technical aspects of app permissions, awareness of permissions described in app stores and the app permission manager and user behavior when encountering suspicious permissions through hypothetical scenarios.

The survey was designed to balance accessibility to smartphone and technical depth or knowledge of the participants. This was done to ensure participants with different technical expertise. Responses were collected over a 30-day period, with $N = 110$ valid records.

3.3 Ethical Considerations

This study, approved by our institutional review board, followed ethical guidelines for human-centered research. Participants provided consent before participating in the survey and their anonymity was assured and maintained. The survey collected no personally identifiable information from the participants and complied with data protection regulations, ensuring the confidentiality of responses.

4 Results

4.1 RQ1: To what extent do Android applications inconsistently disclose permissions between the AndroidManifest.xml file and app descriptions in the Google Play Store?

The analysis of 200 Android applications revealed significant discrepancies between manifest permissions and those disclosed on different Play Store platforms, raising important transparency concerns. Figure 1(b) represents bar charts showing the number of apps categorized by discrepancy percentages for permissions listed on the web and Google Play Store, compared to those in the manifest file.

Mobile Play Store Analysis: From the analysis, it is found that 51% applications (102 out of 200) aligned closely between their manifest permissions and Play Store disclosures, with minimal discrepancies of 0-15%. Another 54 applications displayed moderate discrepancies in the 16-30% range. There were 18 applications with discrepancies of 31-45% and 19 applications with more significant discrepancies of 46-60%. Only 3 applications had severe discrepancies, with 61-75% of their permissions undisclosed. No applications showed discrepancies exceeding 75%.

Web Play Store Analysis: The web-based Play Store interface showed more concerning patterns in permission disclosure. A significant portion of the sample ($N = 87$), had discrepancies of 46-60% between their manifest permissions and web store disclosures. Another 38 applications exhibited even higher discrepancies in the 61-75% range, and 6 applications showed critical discrepancies of 76-90%. Additionally, 50 applications had discrepancies of 31-45%, while 18 applications showed moderate discrepancies of 16-30%. Only one application maintained high transparency with discrepancies under 15%.

4.2 RQ2: How do familiarity, attention, and awareness of app permissions as described in the Google Play Store and managed through the app permission system vary across different demographic groups?

To investigate whether familiarity, attention, and awareness of app permissions vary across demographic groups, statistical analyses were conducted for factors such as age group, education level, and gender. This section summarizes findings on familiarity, attention, and awareness, highlighting relationships with these demographic variables.

Familiarity with App Permissions Across Demographics. A one-way ANOVA was conducted to evaluate whether familiarity with app permissions varies significantly across age groups and education levels. The analysis revealed no statistically significant differences in familiarity across age groups ($F(3, 19.1) = 1.75, p = 0.191$) or education levels ($F(3, 13.3) = 0.959, p = 0.441$). Descriptive statistics in Table 2 indicated that the 25–34 age group had the highest mean familiarity score ($M = 3.88, SD = 1.26$). However, these differences were not statistically significant. From the survey results shown in Figure 2(a), younger participants, particularly those aged 18–24, show the majority (53%) has only low to moderate familiarity (1–3).

The 25–34 age group demonstrates slightly better familiarity, with 76% rating it as high in average. However, familiarity declines further in older age groups, with 35–44 and 45+ participants showing 51% and 59% reporting low to moderate familiarity (1–3), respectively. These findings highlight a general lack of familiarity with app permissions across all age groups. In terms of gender, an independent samples t-test found no significant differences in familiarity scores between male and female participants ($t(108) = 0.929, p = 0.355$). As shown in Figure 2(d), among females, the familiarity lacking is reported as 47% on average (1-3), almost similar to males (49%).

Table 2: Descriptive Statistics for Familiarity with App Permissions by Age Group

Age Group	N	Mean	SD	SE
18–24	65	3.26	1.50	0.186
25–34	24	3.88	1.26	0.258
35–44	6	3.17	1.47	0.601
45 and above	15	2.87	1.68	0.435

Table 3: Descriptive Statistics of Attention given to App Permission Scores by Age Group

Age Group	N	Mean	SD	SE
18–24	65	2.97	1.53	0.190
25–34	24	3.38	1.64	0.334
35–44	6	1.50	1.22	0.500
45 and above	15	2.73	1.58	0.408

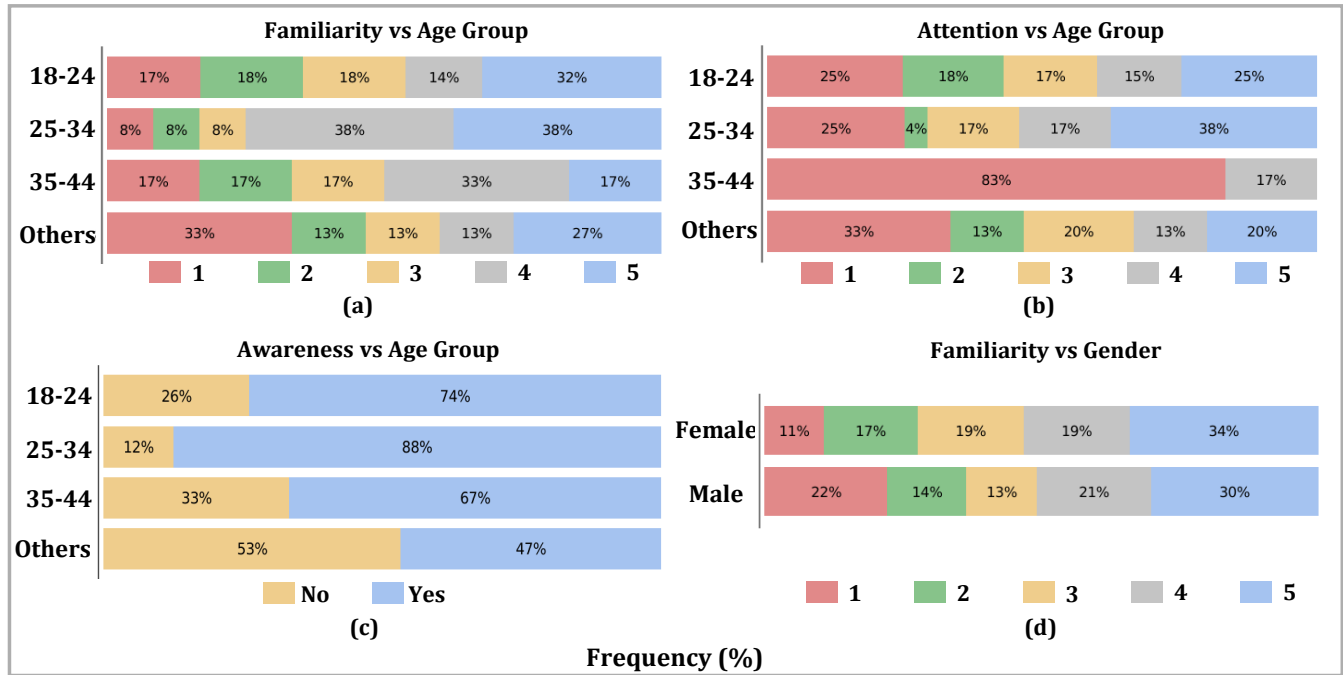


Figure 2: Survey responses of (a) Familiarity vs Age Group (b) Attention vs Age Group (c) Awareness vs Age Group (d) Familiarity vs Gender

Attention Given to App Permissions Across Age Groups. Descriptive statistics in Table 3 indicated that the age group 25 to 34 had the highest mean attention score ($M = 3.38$, $SD = 1.64$). A one-way ANOVA revealed a significant difference in attention given to app permissions while installing an app from app stores across age groups ($F(3, 19.8) = 3.17$, $p = 0.047$). The post hoc Tukey tests showed that the 25-34 age group paid significantly more attention to app permissions than the 35-44 group (mean difference = 1.88, $p = 0.045$). According to the results of the survey shown in Figure 2(b), when it comes to paying attention to the permissions of the app, 60% of the 18-24 age group gives it the lowest to moderate score (1-3). The 25-34 group shows overall better focus, with 55% rating their attention as moderate high to highest (4-5). However, the 35-44 group shows the highest lack of attention, as 83% gave it the lowest rating. These results indicate that while younger people are somewhat attentive, middle-aged people are finding it difficult to pay attention to the app permissions.

Awareness of App Permissions Listed in Play Store Descriptions and Permission Manager Across Demographics. Whether the participants are aware of the consistency of permissions listed in the description of the app store and the permission managers, was analyzed using a chi-square independence test. The results ($\chi^2(10, N = 110) = 32.3$, $p < 0.001$) showed a statistically significant association between age group and awareness. Awareness of app permissions is generally high among younger participants, with 74% of the 18-24 age group and 88% of the 25-34 age group responding positively. This percentage decreases for the 35 to 44 (67%) and even more for the 45+ older age groups (47%). Although younger people are aware of

the app permissions listed in the Play Store app descriptions, this does not always mean consistent familiarity or attention.

4.3 RQ3: How does user behavior vary when encountering unnecessary or mismatched app permissions?

To address RQ3—we analyzed user responses segmented by awareness and age groups of app permissions. User behavior significantly varies when encountering unnecessary or mismatched app permissions, depending on their awareness of app permissions. As shown in Figure 3(a), users who are aware of app permissions are more proactive, as evidenced by their preference for actions such as uninstalling the app immediately (38% respondents) or reporting the issue to the Play Store (24% respondents). Conversely, users who lack awareness tend to exhibit less decisive behavior, with the majority opting to ignore the mismatch (43% respondents). Across the age groups, 18% of 18-24 and 17% of 25-34 age group users chose "Ignore permission mismatch and continue using app". This percentage increases for 35-44 (50%) and 45+ (53%) age group participants. For users with higher awareness levels showed more proactive behaviors, such as "uninstalling the app immediately" or "reporting the issue to the Play Store". For instance, in the 18-24 age group, 28% of aware users chose to uninstall the app, while 23% selected to report it. On the other hand, less aware users in this age group tended to ignore the mismatch or continue using the app.

Chi-square test reveals a significant overall association between awareness and user actions ($\chi^2 = 11.61$, $df = 4$, $p = 0.020$). However, when broken down by age groups, the results are not statistically significant for specific groups ($p > 0.05$ in all cases). The

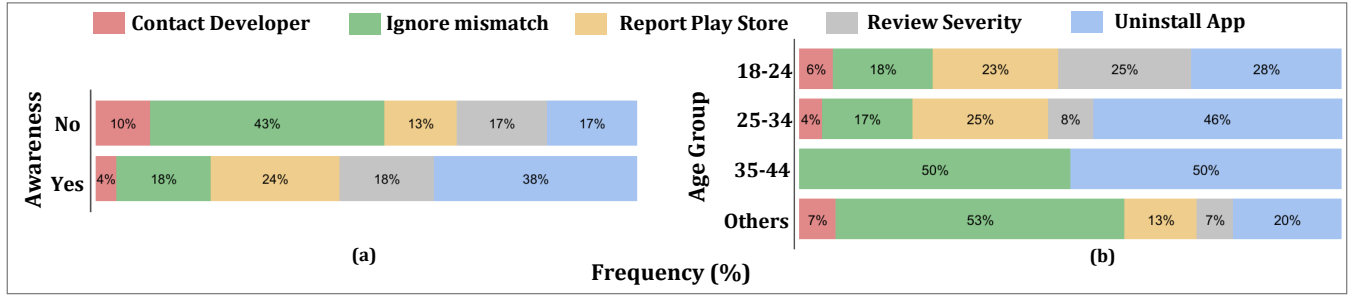


Figure 3: Survey responses of (a) Awareness vs Actions taken (b) Age group vs Actions taken

25–34 and 18–24 age groups exhibited the highest levels of proactive behavior, particularly among aware users. This indicates that while awareness significantly influences user behavior overall, age-specific patterns may vary.

5 Discussion

The findings from *RQ1* highlight a systematic issue in permission transparency on the web platform, emphasizing the need for improved alignment between manifest permissions and web store disclosures. Our analysis of 200 applications reveals that a significant portion of apps exhibit discrepancies between the permissions declared in the app descriptions and those actually requested in the `AndroidManifest.xml` file. This lack of transparency can mislead users, reducing their ability to make informed decisions regarding data privacy. Moreover, the discrepancy is more pronounced in web-based Play Store descriptions, where permission details are often incomplete or absent. To mitigate this issue, app marketplaces should enforce stricter guidelines requiring developers to ensure that all requested permissions are accurately listed in app descriptions. Automated verification tools could be implemented to cross-check declared permissions against manifest files before approval.

From the *RQ2* analysis, there were some significant findings. A significant proportion of users, regardless of their technical proficiency, demonstrate limited familiarity with app permissions. The majority of participants were younger users (aged 18–24, $N = 65$). This group exhibit a mean familiarity score of 3.26, which remains below a highly proficient level and indicates moderate familiarity. Middle-aged users, particularly those aged 35–44, show an even lower level of familiarity, as well as with a decline in attention to permission details during installation. This gap in awareness suggests that only providing permission details is insufficient—better engagement strategies are needed to capture user attention effectively. One potential solution is to redesign permission prompts to be more interactive and context-aware. For example, rather than displaying a list of permissions at the time of installation, developers could implement progressive disclosure techniques that provide explanations at the moment when permissions are being accessed. Enhancing permission interfaces with clearer justifications and visual indicators (e.g., color-coded risk levels) may further help users grasp the implications of granting permissions.

The findings from *RQ3* highlight that users who are aware of app permissions are significantly more likely to take proactive security

actions, such as uninstalling the app or reporting discrepancies. Conversely, users with lower awareness tend to exhibit riskier behaviors, including ignoring mismatches or continuing to use apps without scrutiny. These behaviors indicate a broader need to instill proactive security habits among users, ensuring they critically evaluate permission requests. To address this issue, app ecosystems should integrate proactive educational methods. These could include periodic security reminders that encourage users to review permissions of installed apps. Additionally, operating systems could introduce permission dashboards that provide historical tracking of permission usage, allowing users to evaluate how often an app accesses certain data.

To enhance user trust, both app developers and platform regulators must take responsibility for improving permission disclosures and minimizing discrepancies. From a design perspective, permission systems should be restructured to prioritize user understanding. Strategies such as real-time permission alerts, user-friendly permission settings, and mandatory transparency statements can significantly improve awareness and decision-making. From a policy standpoint, regulatory bodies should enforce standardized permission disclosure formats across all platforms. Developers who fail to comply with transparency guidelines should face stricter penalties, including app removal or restricted distribution. Additionally, user-reported discrepancies should be accounted into app ratings, allowing collective feedback to hold developers accountable. By combining these design improvements and regulatory measures, the app ecosystem can move toward a more transparent and user-centric permission model, ultimately fostering a higher level of trust and security in mobile applications.

6 Conclusion

This study analyzed the discrepancy between the Play Store’s listed app permissions and the app manifest file and explored the familiarity, attention, and awareness of app permissions across different demographic groups, providing insights into user behavior in certain privacy risk situations. It is evident that both young and old age group people irrespective of gender, lack familiarity, attention, and awareness about app permissions. These findings emphasize the need for targeted educational initiatives and user interface designs that can bridge the awareness gap among all age groups. Future research could expand by examining additional demographic variables, such as occupation and cultural background. By identifying gaps in familiarity, attention, and awareness related to app

permissions, this research provides a foundation for enhancing user-centric privacy practices, ensuring that app permission systems provide to diverse user needs and promote informed decision-making in an increasingly digital world.

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Appendices

A Survey Questionnaires

The survey conducted for the user study included the following sections and questions:

A.1 Demographics

These questions (Q1–Q6) aimed to gather information about participants' age, gender, educational background, and app usage frequency.

- Q1: Which age group are you in?
- Q2: What is your gender?
- Q3: What is your country of residence?
- Q4: Which of the following best describes your ethnicity?
- Q5: What is your highest level of education?
- Q6: Please state your area of work or area of study.

A.2 Smartphone Usage

These questions (Q7–Q10) aimed to understand participants' smartphone usage behavior.

- Q7: Do you use a smartphone? If yes, which mobile OS do you use?
- Q8: How much time approximately do you spend using your smartphone daily?

- Q9: Approximately how many times do you try out new apps on your smartphone weekly based on specific factors or features (e.g., recommendations, app reviews, social media trends, or personal needs)?
- Q10: Please choose which of the following apps you use frequently from the list given below (check as many as apply).

A.3 Familiarity, Awareness, and User Behavior on Permissions

Questions (Q11–Q30) assessed users' familiarity with the technical aspects of app permissions, awareness of permissions described in app stores and app permission managers, and user behavior when encountering suspicious permissions through hypothetical scenarios.

- Q11: How familiar are you with app permissions on the Android or iPhone platform?
- Q12: On a scale from 1–5, how much attention do you give to reviewing permissions before granting them?
- Q13: How would you rate your understanding of different types of app permissions requested after installing an app?
- Q14: Are you aware of the features, functionalities, and permissions included in app descriptions on app stores?
- Q15: If yes, do you typically read app permissions in the app description before or after installing an app?
- Q16: Have you ever installed an app that requested access to data/features unnecessary for its functionality (e.g., a flash-light app requesting contact access)?
- Q17: If yes, how likely are you to review the app permissions in the app description before installing it in the future?
- Q18: Have you ever noticed mismatches between permissions listed in the app's description and permissions in the App Manager?
- Q19: Have you experienced apps requesting new permissions after updates? If yes, do you check whether these permissions are listed in the app description?
- Q20: What would you do if permissions in the app description and the App Manager do not match?
- Q21: How would such mismatches influence your trust in the app and future app downloads?

Read the following story and answer the questions: "Uncle Sam recently downloaded a weather app to get daily forecasts and severe weather alerts. After installing the app, he noticed it was requesting access to the phone's location, camera, and microphone. While the location access seemed necessary to provide local weather updates, he couldn't understand why the app needed access to the camera and microphone. But he didn't bother much about the permissions and continued using the app."

- Q22: On a scale from 1–5, how justified do you think it is for a weather app to request access to permissions like the camera or microphone?
- Q23: "Uncle Sam didn't pay much attention to unrelated app permissions and continued using it." To what extent do you agree or disagree?
- Q24: Would you review the weather app permissions after noticing unrelated permissions?

Read the following story and answer the questions: "Suppose you recently installed a third-party cache clear app to clear cache files and background apps to increase the performance. After using the app for a few days, you started noticing personalized ads based on places she had visited. During the app installation, you didn't notice which permissions you gave consent to, so you checked the app permissions from the Android App Manager, and you noticed the app is using location in app permission, but you can't remember giving the app permission to use your location."

- Q25: What would be your next course of action in this situation?
- Q26: On a scale from 1–5, how would you feel if you noticed the app you installed was using your location without listing it in the description?
- Q27: "Jamie noticed a mismatch in the cache clear app but did nothing and continued using it." To what extent do you agree or disagree?